DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC YEAR: 2025-26 • SEMESTER: V

Curriculum Scheme: NEP 2020 / AICTE Model Curriculum • Outcome-Based Education (OBE) Format

Course Name:	**Cloud-Native Distributed Systems and Microservices**		
Course Code:	**22CS501**	**Regulation / Scheme:**	NEP 2020 / AICTE Model Curriculum
Credits (L-T-P-C):	3 - 0 - 2 - 4	**Total Contact Hours:**	**45 Hours**
Theory Hours:	45 Hours	**Practical/Lab Hours:**	30 Hours
Self-Study / Project:	15 Hours	**CIE / SEE Marks:**	50 / 50 (Total: 100)

1. Course Outcomes (COs) & Bloom's Cognitive Mapping

CO ID	Course Outcome Statement	Revised Bloom's Level	Target Attainment
CO1	Understand and explain foundational concepts, formal representations, and system invariants. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L2: Understand**	70.0%
CO2	Apply algorithmic techniques, mathematical models, and architectural patterns to solve domain problems. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L3: Apply**	70.0%
CO3	Analyze system performance, bottlenecks, failure modes, and trade-offs under varying operational constraints. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L4: Analyze**	70.0%
CO4	Formulate and analyze modern data workflows, streaming architectures, and distributed coordination. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L4: Analyze**	70.0%
CO5	Evaluate enterprise integration strategies, security postures, and cost-performance trade-offs. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L5: Evaluate**	70.0%
CO6	Design and synthesize an end-to-end, production-grade industry-aligned system integrating cutting-edge tooling. and formulate engineering solutions for Cloud-Native Distributed Systems and Microservices.	**L6: Create**	70.0%

Bloom's Taxonomy Levels: L1: Remember, L2: Understand, L3: Apply, L4: Analyze, L5: Evaluate, L6: Create.

2. CO-PO and CO-PSO Articulation Matrix (Course Articulation)

Correlation Levels: 3: Substantial (High), 2: Moderate (Medium), 1: Slight (Low), -: No Correlation.

3. Detailed Unit / Module-Wise Syllabus

4. Assessment Scheme and Examination Pattern

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	2	2	2	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2	2	2	2
CO3	3	3	-	2	3	-	-	-	-	-	-	2	2	2	2
CO4	3	3	2	2	3	-	-	-	-	-	-	2	3	3	3
CO5	3	3	2	2	3	-	-	-	-	-	-	2	3	3	3
CO6	3	3	3	-	3	-	-	-	2	2	-	2	3	3	3
Average	**3.0**	**2.8**	**2.3**	**2.0**	**3.0**	**-**	**-**	**-**	**2.0**	**2.0**	**-**	**2.0**	**2.5**	**2.5**	**2.5**

Module 1: Foundational Concepts and Theoretical Framework

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L1-L2

- Introduction to Cloud-Native Distributed Systems and Microservices and core domain principles
- Formal models, architectural taxonomies, and invariant properties
- Mathematical foundations, complexity bounds, and state representation
- Design space trade-offs, consistency criteria, and boundary conditions

Module 2: Core Algorithms and Design Patterns

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L2-L3

- Fundamental algorithms, control flows, and scheduling mechanisms
- Component interactions, interface contracts, and encapsulation
- State synchronization, resource management, and partition recovery
- Analytical verification and baseline validation benchmarks

Module 3: System Architecture and Optimization

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L3-L4

- Modular decomposition, layering, and concurrency abstractions
- High-performance caching, replication, and data partitioning
- Fault-tolerant consensus, failure detectors, and self-healing topologies
- Empirical profiling, bottleneck detection, and latency mitigation

5. Textbooks and Reference Books (IEEE Academic Format)

Textbooks:

1. Martin Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media, 2023. [ISBN: 978-1449373320]
2. Andrew S. Tanenbaum and Maarten van Steen, *Distributed Systems: Principles and Paradigms*, 4th Edition, Pearson Education, 2024. [ISBN: 978-1543057386]

Reference Books & Web Resources:

1. Brendan Burns, *Designing Distributed Systems: Patterns and Paradigms for Scalable, Reliable Services*,

Module 4: Modern Engineering Workflows & Edge AI	**Contact: 9 Hours**	**Weightage: 20.0%**

Cognitive Focus: L4-L5 **Industry Integration:** Edge AI, vLLM

- Integration of Edge AI into modern operational pipelines
- Asynchronous event processing using vLLM and message brokers
- Distributed telemetry, observability, and structured logging
- Zero-trust security, containerized deployment, and CI/CD pipelines

Module 5: Advanced Industry Applications & TensorRT-LLM	**Contact: 9 Hours**	**Weightage: 20.0%**

Cognitive Focus: L5-L6 **Industry Integration:** TensorRT-LLM, Kafka Streams

- Edge computing architectures, TinyML, and hardware acceleration with TensorRT-LLM
- High-throughput LLM serving and model optimization utilizing Kafka Streams
- Fault injection testing, chaos engineering, and resilience metrics
- Scalability benchmarking and compliance with modern enterprise standards

Component	Max Marks	Assessment Mechanism & Rubrics
Continuous Internal Evaluation (CIE)	**50 Marks**	2 Mid-Term Tests (30 Marks) + Quizzes, Problem Sets, and Real-Time Lab Audits (20 Marks).
Semester End Examination (SEE)	**50 Marks**	Comprehensive 3-hour theory examination covering all 5 modules with balanced Bloom's level distribution.
Total Course Marks	**100 Marks**	*Continuous Internal Evaluation (CIE) consists of 2 written tests (30 Marks) and hands-on assignments/quizzes (20 Marks). Semester End Examination (SEE) is a 3-hour written comprehensive evaluation carrying 50 Marks with questions from all 5 modules.*

2nd Edition, O'Reilly Media, 2024. [ISBN: 978-1492031772]

2. IEEE / ACM Joint Curriculum Committee, *Computer Science Curricula 2023: Guidelines for Associate and Baccalaureate Programs in Computing*, Final Edition, IEEE-CS & ACM Press, 2024.

Course Coordinator	**Module Reviewer / HOD**	**Dean (Academics)**
Faculty In-Charge	*Department of Computer Science & Engineering*	*Board of Studies Approval*